

# Lecture 2

## Multiple Linear Regression: Estimation and Inference

Reading: Faraway 2014 Chapters 2.1 - 2.6, 3.1 - 3.2; 3.5; ISLR  
2021 Chapter 3.2

*STAT 8020 Statistical Methods II*

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# Agenda

Multiple Linear  
Regression:  
Estimation and  
Inference



Multiple Linear  
Regression

Estimation & Inference

Assessing Model Fit

**1 Multiple Linear Regression**

**2 Estimation & Inference**

**3 Assessing Model Fit**

## Multiple Linear Regression (MLR)

**Goal:** To model the relationship between two or more predictors ( $x$ 's) and a response ( $y$ ) by fitting a **linear equation** to observed data  $\{y_i, x_{1,i}, x_{2,i}, \dots, x_{p-1,i}\}_{i=1}^n$ :

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \dots + \beta_{p-1} x_{p-1,i} + \varepsilon_i, \quad \varepsilon_i \stackrel{i.i.d.}{\sim} N(0, \sigma^2)$$

**Example:** Species diversity on the Galapagos Islands. We are interested in studying the relationship between the number of plant species (Species) and the following geographic variables: Area, Elevation, Nearest, Scruz, Adjacent.



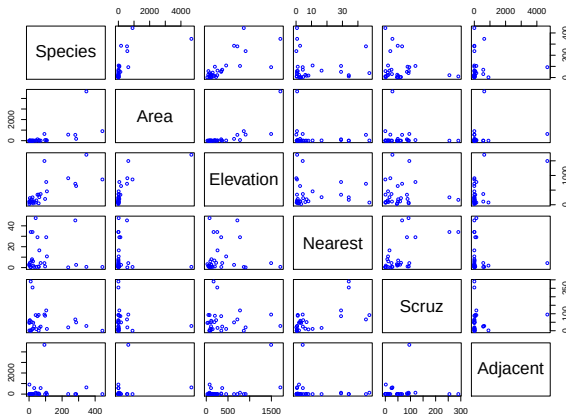
**Key idea:** Real-world data are multivariate – multiple drivers affect the response

# Data: Species Diversity on the Galapagos Islands

|              | Species | Endemics | Area    | Elevation | Nearest | Scruz | Adjacent |
|--------------|---------|----------|---------|-----------|---------|-------|----------|
| Baltra       | 58      | 23       | 25.09   | 346       | 0.6     | 0.6   | 1.84     |
| Bartolome    | 31      | 21       | 1.24    | 109       | 0.6     | 26.3  | 572.33   |
| Caldwell     | 3       | 3        | 0.21    | 114       | 2.8     | 58.7  | 0.78     |
| Champion     | 25      | 9        | 0.10    | 46        | 1.9     | 47.4  | 0.18     |
| Coamano      | 2       | 1        | 0.05    | 77        | 1.9     | 1.9   | 903.82   |
| Daphne.Major | 18      | 11       | 0.34    | 119       | 8.0     | 8.0   | 1.84     |
| Daphne.Minor | 24      | 0        | 0.08    | 93        | 6.0     | 12.0  | 0.34     |
| Darwin       | 10      | 7        | 2.33    | 168       | 34.1    | 290.2 | 2.85     |
| Eden         | 8       | 4        | 0.03    | 71        | 0.4     | 0.4   | 17.95    |
| Enderby      | 2       | 2        | 0.18    | 112       | 2.6     | 50.2  | 0.10     |
| Espanola     | 97      | 26       | 58.27   | 198       | 1.1     | 88.3  | 0.57     |
| Fernandina   | 93      | 35       | 634.49  | 1494      | 4.3     | 95.3  | 4669.32  |
| Gardner1     | 58      | 17       | 0.57    | 49        | 1.1     | 93.1  | 58.27    |
| Gardner2     | 5       | 4        | 0.78    | 227       | 4.6     | 62.2  | 0.21     |
| Genovesa     | 40      | 19       | 17.35   | 76        | 47.4    | 92.2  | 129.49   |
| Isabela      | 347     | 89       | 4669.32 | 1707      | 0.7     | 28.1  | 634.49   |
| Marchena     | 51      | 23       | 129.49  | 343       | 29.1    | 85.9  | 59.56    |
| Onslow       | 2       | 2        | 0.01    | 25        | 3.3     | 45.9  | 0.10     |
| Pinta        | 104     | 37       | 59.56   | 777       | 29.1    | 119.6 | 129.49   |
| Pinzon       | 108     | 33       | 17.95   | 458       | 10.7    | 10.7  | 0.03     |
| Las.Plazas   | 12      | 9        | 0.23    | 94        | 0.5     | 0.6   | 25.09    |
| Rabida       | 70      | 30       | 4.89    | 367       | 4.4     | 24.4  | 572.33   |
| SanCristobal | 280     | 65       | 551.62  | 716       | 45.2    | 66.6  | 0.57     |
| SanSalvador  | 237     | 81       | 572.33  | 906       | 0.2     | 19.8  | 4.89     |
| SantaCruz    | 444     | 95       | 903.82  | 864       | 0.6     | 0.0   | 0.52     |
| SantaFe      | 62      | 28       | 24.08   | 259       | 16.5    | 16.5  | 0.52     |
| SantaMaria   | 285     | 73       | 170.92  | 640       | 2.6     | 49.2  | 0.10     |
| Seymour      | 44      | 16       | 1.84    | 147       | 0.6     | 9.6   | 25.09    |
| Tortuga      | 16      | 8        | 1.24    | 186       | 6.8     | 50.9  | 17.95    |
| Wolf         | 21      | 12       | 2.85    | 253       | 34.1    | 254.7 | 2.33     |

**Observation:** predictors vary on very different scales

# How Do Geographic Variables Affect Species Diversity?



**Look for:**

- Linear vs nonlinear relationships
- Strength of association
- Potential outliers

## Let's Take a Look at the Correlation Matrix

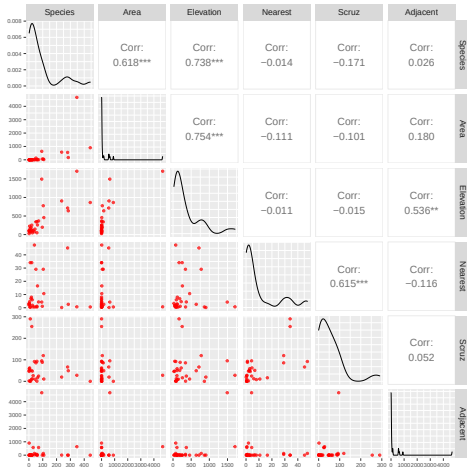
Here we compute the correlation coefficients between the response (`Species`) and predictors (all the geographic variables)

```
> round(cor(gala[, -2]), 3)
```

|           | Species | Area   | Elevation | Nearest | Scruz  | Adjacent |
|-----------|---------|--------|-----------|---------|--------|----------|
| Species   | 1.000   | 0.618  | 0.738     | -0.014  | -0.171 | 0.026    |
| Area      | 0.618   | 1.000  | 0.754     | -0.111  | -0.101 | 0.180    |
| Elevation | 0.738   | 0.754  | 1.000     | -0.011  | -0.015 | 0.536    |
| Nearest   | -0.014  | -0.111 | -0.011    | 1.000   | 0.615  | -0.116   |
| Scruz     | -0.171  | -0.101 | -0.015    | 0.615   | 1.000  | 0.052    |
| Adjacent  | 0.026   | 0.180  | 0.536     | -0.116  | 0.052  | 1.000    |

**Key takeaway:** Correlation captures only linear association and may fail to reflect nonlinear patterns in the data

# Combining Two Pieces of Information in One Plot



## Interpretation:

- Diagonal: distribution of each variable
- Upper: correlation values
- Lower: scatter relationships

## Model 1: Species ~ Elevation

```
Call:
lm(formula = Species ~ Elevation, data = gala)

Residuals:
    Min       1Q   Median       3Q      Max
-218.319  -30.721  -14.690    4.634   259.180

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  11.33511    19.20529   0.590    0.56
Elevation     0.20079     0.03465   5.795 3.18e-06 ***
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

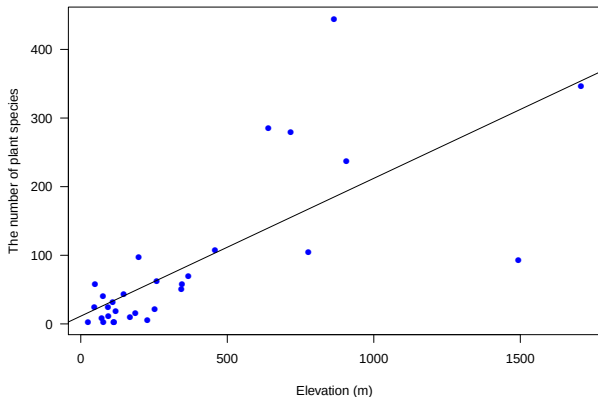
Residual standard error: 78.66 on 28 degrees of freedom
Multiple R-squared:  0.5454,    Adjusted R-squared:  0.5291
F-statistic: 33.59 on 1 and 28 DF,  p-value: 3.177e-06
```

This is the familiar simple linear regression output



## Model 1 Fit

$$\hat{\text{Species}} = 11.33511 + 0.20079 \times \text{Elevation},$$
$$\hat{\sigma} = 78.66, R^2 = 0.5454$$



**Key observation:** positive relationship, but substantial variability remains

## Model 2: Species ~ Elevation + Area

```
Call:
lm(formula = Species ~ Elevation + Area, data = gala)
```

Residuals:

| Min      | 1Q      | Median  | 3Q    | Max     |
|----------|---------|---------|-------|---------|
| -192.619 | -33.534 | -19.199 | 7.541 | 261.514 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | 17.10519 | 20.94211   | 0.817   | 0.42120    |
| Elevation   | 0.17174  | 0.05317    | 3.230   | 0.00325 ** |
| Area        | 0.01880  | 0.02594    | 0.725   | 0.47478    |

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Signif. codes:

0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 79.34 on 27 degrees of freedom

Multiple R-squared: 0.554, Adjusted R-squared: 0.521

F-statistic: 16.77 on 2 and 27 DF, p-value: 1.843e-05

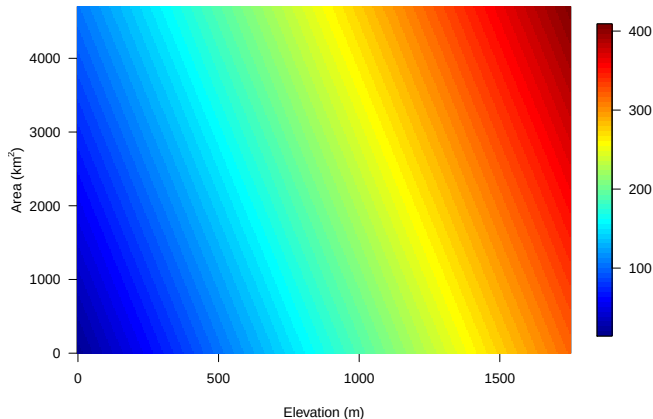
This looks similar to simple linear regression output, but now includes two predictors

## Model 2 Fit

$$\hat{\text{Species}} = 17.10519 + 0.17174 \times \text{Elevation} + 0.01880 \times \text{Area},$$

$$\hat{\sigma} = 79.34, R^2 = 0.554$$

**Interpretation:** predicted species count as a (linear) function of Elevation and Area



## Model 3: Species ~ Elevation + Area + Adjacent

```
Call:
lm(formula = Species ~ Elevation + Area + Adjacent, data = gala)
```

Residuals:

| Min      | 1Q      | Median | 3Q     | Max     |
|----------|---------|--------|--------|---------|
| -124.064 | -34.283 | -8.733 | 27.972 | 195.973 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )    |
|-------------|----------|------------|---------|-------------|
| (Intercept) | -5.71893 | 16.90706   | -0.338  | 0.73789     |
| Elevation   | 0.31498  | 0.05211    | 6.044   | 2.2e-06 *** |
| Area        | -0.02031 | 0.02181    | -0.931  | 0.36034     |
| Adjacent    | -0.07528 | 0.01698    | -4.434  | 0.00015 *** |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 61.01 on 26 degrees of freedom

Multiple R-squared: 0.746, Adjusted R-squared: 0.7167

F-statistic: 25.46 on 3 and 26 DF, p-value: 6.683e-08

- Which predictors are significant?
- How do coefficients change?

## “Full Model”

```
lm(formula = Species ~ Area + Elevation + Nearest + Scruz + Adjacent,  
    data = gala)
```

Residuals:

|  | Min      | 1Q      | Median | 3Q     | Max     |
|--|----------|---------|--------|--------|---------|
|  | -111.679 | -34.898 | -7.862 | 33.460 | 182.584 |

Coefficients:

|             | Estimate  | Std. Error | t value | Pr(> t ) |
|-------------|-----------|------------|---------|----------|
| (Intercept) | 7.068221  | 19.154198  | 0.369   | 0.715351 |
| Area        | -0.023938 | 0.022422   | -1.068  | 0.296318 |
| Elevation   | 0.319465  | 0.053663   | 5.953   | 3.82e-06 |
| Nearest     | 0.009144  | 1.054136   | 0.009   | 0.993151 |
| Scruz       | -0.240524 | 0.215402   | -1.117  | 0.275208 |
| Adjacent    | -0.074805 | 0.017700   | -4.226  | 0.000297 |

(Intercept)

Area

Elevation \*\*\*

Nearest

Scruz

Adjacent \*\*\*

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Signif. codes:

0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 60.98 on 24 degrees of freedom

Multiple R-squared: 0.7658, Adjusted R-squared: 0.7171

F-statistic: 15.7 on 5 and 24 DF, p-value: 6.838e-07

Do all predictors explain response variability?

Similar to SLR, we will discuss

- Estimation
- Inference
- Diagnostics and Remedies

We will also discuss some new topics

- Model Selection
- Multicollinearity

## Multiple Linear Regression in Matrix Notation

Given the actual data, we can write MLR model as:

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_{1,1} & x_{2,1} & \cdots & x_{p-1,1} \\ 1 & x_{1,2} & x_{2,2} & \cdots & x_{p-1,2} \\ \vdots & \cdots & \ddots & \vdots & \vdots \\ 1 & x_{1,n} & x_{2,n} & \cdots & x_{p-1,n} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_{p-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

It will be more convenient to put this in a matrix representation as:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Error Sum of Squares (SSE) =  $\sum_{i=1}^n (y_i - (\beta_0 + \sum_{j=1}^{p-1} \beta_j x_{j,i}))^2$   
can be expressed as:

$$(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

Next, we are going to find  $\hat{\boldsymbol{\beta}} = (\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_{p-1})$  to minimize SSE as our estimate for  $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_{p-1})$

## Estimating Regression Coefficients

We apply method of least squares to minimize  $(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$  to obtain  $\hat{\boldsymbol{\beta}}$

- The resulting least squares estimate is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

(see LS\_MLR.pdf for the derivation)

- Fitted values:

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \mathbf{H}\mathbf{y}$$

- Residuals:

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\mathbf{y}$$



- Similar as we did in SLR

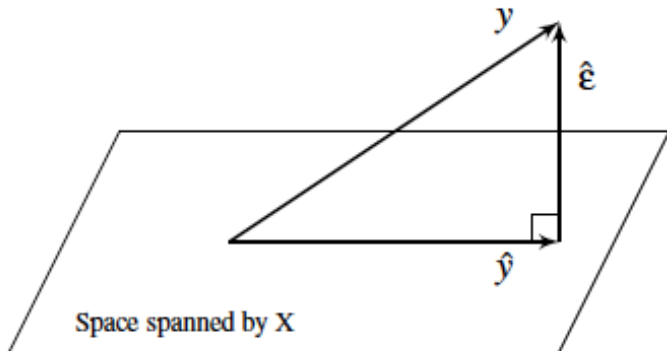
$$\begin{aligned}\hat{\sigma}^2 &= \frac{\mathbf{e}^T \mathbf{e}}{n - p} \\ &= \frac{(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})^T (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})}{n - p} \\ &= \frac{\text{SSE}}{n - p} \\ &= \text{MSE}\end{aligned}$$

**Mean Squared Error (MSE):** average squared residual

**Residual standard error (R output):**  $\hat{\sigma} = \sqrt{\text{MSE}}$

## Geometrical Representation of the Estimation $\beta$

Projecting the observed response  $y$  into a space spanned by  $X$



Source: Linear Model with R 2nd Ed, Faraway, p. 15

**Key idea:** we project data onto the space spanned by predictors

## Partitioning Sums of Squares

- Total sums of squares in response

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

- We can rewrite SST as

$$\begin{aligned} \sum_{i=1}^n (y_i - \bar{y})^2 &= \sum_{i=1}^n (y_i - \hat{y}_i + \hat{y}_i - \bar{y})^2 \\ &= \underbrace{\sum_{i=1}^n (y_i - \hat{y}_i)^2}_{\text{"Error": SSE}} + \underbrace{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}_{\text{Model: SSR}} \end{aligned}$$

# Partitioning Total Sums of Squares: A Graphical Illustration

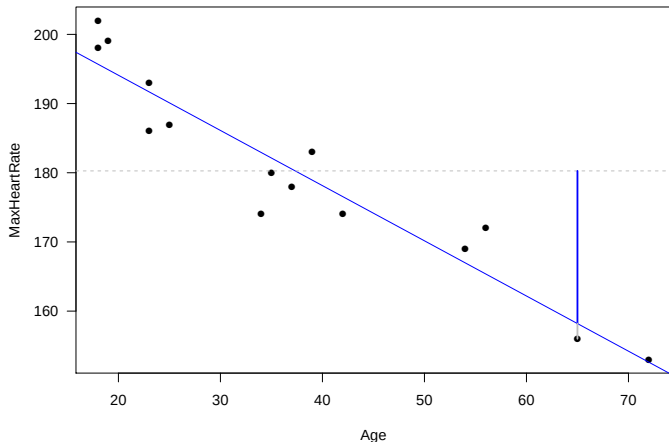
Multiple Linear  
Regression:  
Estimation and  
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Multiple Linear  
Regression

Estimation & Inference

Assessing Model Fit



Total variation = explained (blue) + unexplained (gray)

## ANOVA Table & $F$ -Test

To answer the question: **Is at least one of the predictors  $x_1, \dots, x_{p-1}$  useful in predicting the response  $y$ ?**

| Source | df      | SS  | MS                  | F Value   |
|--------|---------|-----|---------------------|-----------|
| Model  | $p - 1$ | SSR | $MSR = SSR/(p - 1)$ | $MSR/MSE$ |
| Error  | $n - p$ | SSE | $MSE = SSE/(n - p)$ |           |
| Total  | $n - 1$ | SST |                     |           |

- $F$ -Test: Tests if the predictors  $\{x_1, \dots, x_{p-1}\}$  collectively help explain the variation in  $y$

- $H_0 : \beta_1 = \beta_2 = \dots = \beta_{p-1} = 0$
- $H_a : \text{at least one } \beta_k \neq 0, \quad 1 \leq k \leq p - 1$
- $F^* = \frac{MSR}{MSE} = \frac{SSR/(p-1)}{SSE/(n-p)} \stackrel{H_0}{\sim} F_{p-1, n-p}$
- Reject  $H_0$  if  $F^* > F_{1-\alpha, p-1, n-p}$

# Testing Individual Predictor

- We can show that  $\hat{\beta} \sim N_p(\beta, \sigma^2 (X^T X)^{-1}) \Rightarrow$   
 $\hat{\beta}_k \sim N(\beta_k, \sigma_{\hat{\beta}_k}^2)$

- Perform  $t$ -Test:

- $H_0 : \beta_k = 0$  vs.  $H_a : \beta_k \neq 0$
- $\frac{\hat{\beta}_k - \beta_k}{SE(\hat{\beta}_k)} \sim t_{n-p} \Rightarrow t^* = \frac{\hat{\beta}_k}{SE(\hat{\beta}_k)} \stackrel{H_0}{\sim} t_{n-p}$
- Reject  $H_0$  if  $|t^*| > t_{1-\alpha/2, n-p}$

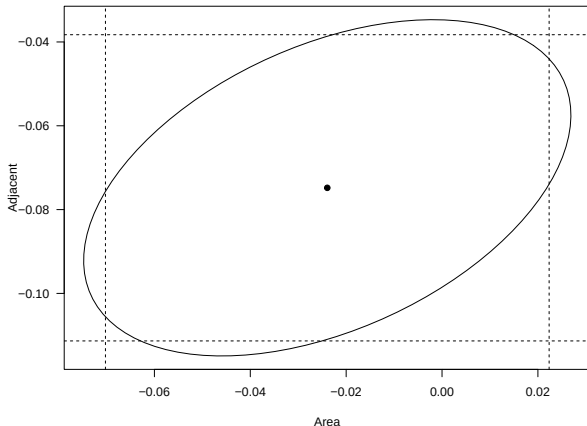
- Confidence interval for  $\beta_k$ :

$$\hat{\beta}_k \pm t_{1-\alpha/2, n-p} SE(\hat{\beta}_k)$$

## Confidence Intervals and Confidence Ellipsoids

Comparing with individual confidence interval, confidence ellipsoids can provide additional information when inference with multiple parameters is of interest. A  $100(1 - \alpha)\%$  confidence ellipsoid for  $\beta$  can be constructed using:

$$(\hat{\beta} - \beta)^T \mathbf{X}^T \mathbf{X} (\hat{\beta} - \beta) \leq p \hat{\sigma}^2 F_{p, n-p}^{\alpha}.$$



## Quantifying Model Fit using Coefficient of Determination $R^2$

- Coefficient of determination  $R^2$  describes proportional of the variance in the response variable that is predictable from the predictors

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}, \quad 0 \leq R^2 \leq 1$$

- $R^2$  increases with the increasing  $p$ , the number of the predictors
  - Adjusted  $R^2$ , denoted by  $R_{\text{adj}}^2 = 1 - \frac{SSE/(n-p)}{SST/(n-1)}$  attempts to account for  $p$



Suppose the true relationship between response  $y$  and predictors  $(x_1, x_2)$  is

$$y = 5 + 2x_1 + \varepsilon,$$

where  $\varepsilon \sim N(0, 1)$  and  $x_1$  and  $x_2$  are independent to each other. Let's fit the following two models to the "data"

$$\text{Model 1: } y = \beta_0 + \beta_1 x_1 + \varepsilon^1$$

$$\text{Model 2: } y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon^2$$

**Question:** Which model will "win" in terms of  $R^2$ ?

Let's conduct a **Monte Carlo** simulation to study this

- 1 Generating a large number (e.g.,  $M = 500$ ) of “data sets”, where each has exactly the same  $\{x_{1,i}, x_{2,i}\}_{i=1}^n$  but different values of response  $\{y_i = 5 + 2x_{1,i} + \varepsilon_i\}_{i=1}^n$
- 2 Fitting model 1:  $y = \beta_0 + \beta_1 x_1 + \varepsilon^1$  (true model) and model 2:  $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon^2$ , respectively for each simulating data set and calculating their  $R^2$  and  $R_{adj}^2$
- 3 Summarizing  $\{R_j^2\}_{j=1}^M$  and  $\{R_{adj,j}^2\}_{j=1}^M$  for model 1 and model 2

# An Example of Model 1 Fit

```
> summary(fit1)
```

Call:

```
lm(formula = y ~ x1)
```

Residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -1.6085 | -0.5056 | -0.2152 | 0.6932 | 2.0118 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | 5.1720   | 0.1534     | 33.71   | < 2e-16 ***  |
| x1          | 1.8660   | 0.1589     | 11.74   | 2.47e-12 *** |

---

Signif. codes:

0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8393 on 28 degrees of freedom

Multiple R-squared: 0.8313, Adjusted R-squared: 0.8253

F-statistic: 138 on 1 and 28 DF, p-value: 2.467e-12

## An Example of Model 2 Fit

```
> summary(fit2)
```

Call:

```
lm(formula = y ~ x1 + x2)
```

Residuals:

|  | Min     | 1Q      | Median  | 3Q     | Max    |
|--|---------|---------|---------|--------|--------|
|  | -1.3926 | -0.5775 | -0.1383 | 0.5229 | 1.8385 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | 5.1792   | 0.1518     | 34.109  | < 2e-16 ***  |
| x1          | 1.8994   | 0.1593     | 11.923  | 2.88e-12 *** |
| x2          | -0.2289  | 0.1797     | -1.274  | 0.213        |

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Signif. codes:

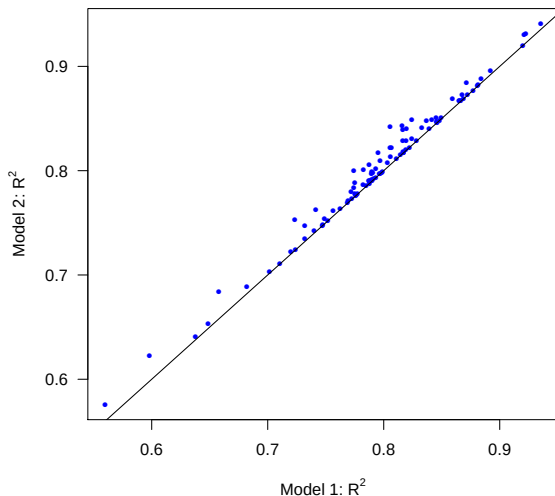
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Residual standard error: 0.8301 on 27 degrees of freedom

Multiple R-squared: 0.8408, Adjusted R-squared: 0.8291

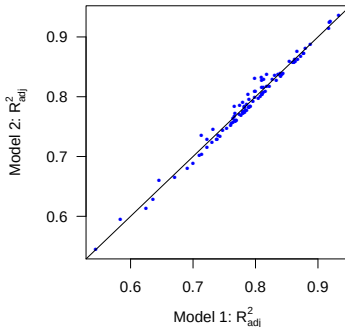
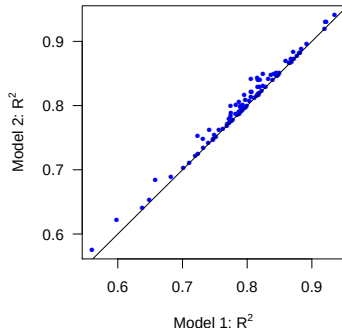
F-statistic: 71.32 on 2 and 27 DF, p-value: 1.677e-11

## $R^2$ : Model 1 vs. Model 2



**Key observation:** Model 2 almost always has higher  $R^2$

## $R_{adj}^2$ : Model 1 vs. Model 2



### Takeaways:

- $R^2$  always pick the more “complex” model (i.e., with more predictors), even the simpler model is the true model
- $R_{adj}^2$  has a better chance to pick the “right” model

# Summary

These slides cover:

- Parameter Estimation of MLR
- Inference: F-test and t-test; Confidence intervals/ellipsoids
- Assessing Model Fit:  $R^2$  and  $R^2_{\text{adj}}$
- Monte Carlo Simulation

R functions to know:

- `image.plot` in the `fields` library and `scatter3D` in the `plot3D` library for visualization
- `anova` for computing the ANOVA table